

SAHIL KUMAR

Generative AI Engineer | RAG · Agentic AI · LangGraph · LLM Evaluation · FastAPI

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India · Open to relocation and remote · [Portfolio](#)

PROFESSIONAL SUMMARY

Generative AI Engineer with **4 years of experience**, including **2.5+ years in GenAI** delivering RAG pipelines, multi-agent workflows and FastAPI backends in Python for client teams. I build for reliability: claim-level hallucination checks, schema-validated outputs, evaluation harnesses and automated tests, backed by an L2 support and data background.

TECHNICAL SKILLS

GenAI & LLMs: Prompt engineering, structured outputs, tool calling, guardrails, LoRA/QLoRA, OpenAI, Azure OpenAI, Groq
RAG: LlamaIndex, FAISS, ChromaDB, Pinecone, pgvector, embeddings, hybrid search (BM25 + vector), reranking, chunking
Agentic AI: LangGraph, LangChain, CrewAI, MCP, multi-agent workflows, tool use, human-in-the-loop
Evaluation & Observability: RAGAS, DeepEval, LangSmith, Langfuse, Recall@k/MRR, hallucination detection (NLI)
Backend & Cloud: Python, Node.js, FastAPI, Pydantic, Docker, GitHub Actions, AWS, Azure, pytest
Data & Messaging: SQL, PostgreSQL, NoSQL, MongoDB, Redis, Kafka
ML: Hugging Face Transformers, Ollama, scikit-learn, XGBoost, Pandas

EXPERIENCE

Generative AI Engineer · Freelance

Mar 2024 – Present

Client code under NDA; architecture walkthroughs available on request.

- Delivered **20+ GenAI projects for 10+ clients** (India, US, Europe) in HR, customer support, e-commerce and real estate.
- Built and debugged RAG pipelines (LangChain, ChromaDB), chatbots, and text-to-SQL for e-commerce analytics.
- Set up RAGAS evaluation (faithfulness, answer relevance, context precision) to tune chunking and retrieval.
- Automated multi-step internal workflows with LangGraph and CrewAI agents.
- Lowered LLM inference cost through prompt compression, embedding caching and tighter context windows.
- Containerized services with Docker and deployed them on AWS and Azure with GitHub Actions CI/CD.

Technical Process Associate – L2 Outbound · Tech Mahindra

Jun 2023 – Dec 2024

- Resolved escalated Level 2 cases through structured, step-by-step troubleshooting under tight turnaround times.
- Wrote and maintained process documentation for recurring issues.

Data Analyst · Freelance

Aug 2022 – Oct 2023

- Cleaned and analysed business datasets with Python, Pandas and SQL, and built dashboards and insight reports.

PROJECTS

3GPP Spec RAG Chatbot with Hallucination Control | FastAPI, FAISS, Cross-Encoder, NLI, Groq [Live Demo](#) · [GitHub](#)

- Built Q&A over synthetic 3GPP-style specs: retrieve → rerank → generate → NLI-verify, with clause citations.
- 74-question held-out set: **93% answer coverage, 0% hallucination** on adversarial cases; MRR 0.78 → 0.85.
- Debugged NLI verifier false refusals (claim cleaning, sentence-level matching): **wrong refusals 44% → 11%**; 49 tests.

RateCon Extract – LLM Document Extraction | Python, OpenAI Structured Outputs, Pydantic, pytest

[GitHub](#)

- Built an LLM pipeline that turns unstructured freight rate confirmations into schema-validated JSON.
- Held-out set of 16 documents: **0 wrong fields out of 192**, composite score 0.92, no wrong high-confidence auto-fills.
- Validation checks feed a confidence score routing results to auto-fill, review or reject; 58 tests incl. mock-LLM failures.

Multi-Document AI Assistant (RAG) | LangChain, ChromaDB, Hugging Face, Groq, Streamlit

[Live Demo](#) · [GitHub](#)

- Built grounded RAG over PDFs with a history-aware MMR retriever, page-level citations and streaming.

AI Multi-Agent System Designer | Python, CrewAI, LangGraph, OpenRouter, Streamlit

[Live Demo](#) · [GitHub](#)

- Orchestrated 4 CrewAI agents with LangGraph to turn product ideas into requirements, architecture and API design.

CERTIFICATIONS

- Model Context Protocol: Advanced Topics** · Anthropic
- Agents and Workflows** · OpenAI
- Claude Code in Action** · Anthropic
- SQL (Advanced)** · HackerRank

EDUCATION

Bachelor of Commerce (BCom) · Maharaja Agrasen Himalayan Garhwal University, Uttarakhand

2019 – 2022

Senior Secondary (CBSE) · Delhi Public School, Roorkee

2019